

Req ID	Requirement Description	Code Mapping (Module/Function)	UT Mapping	Logical IT Mapping	Physical IT Mapping	Status
ECSBF_FR_01	Node-Engine secure handshake	Client::connectToEngine(), ECSBF_CoreEngine::startEngine()	UT_NetworkUtil (sendPacket/receivePacket)	LIT-SUN-01, LIT-SUN-03	PIT-SUN-01	Covered
ECSBF_FR_02	Node provisioning before access	IdentityRegistry::registerNode()	UT-SUN-01, UT-SUN-03	LIT-SUN-01	PIT-SUN-02	Covered
ECSBF_FR_03	Mandatory identity verification	IdentityRegistry::authenticate()	UT-SUN-02, UT-RAIN-01, UT-RAIN-02	LIT-SUN-04, LIT-RAIN-03, LIT-RAIN-04	PIT-SUN-03, PIT-RAIN-01, PIT-RAIN-02	Covered
ECSBF_FR_04	Persistent session management	Session (create, restore, invalidate)	UT-SUN-04, UT-SUN-05, UT-SUN-06, UT-RAIN-03, UT-RAIN-04	LIT-SUN-05, LIT-SUN-13, LIT-SUN-14	PIT-SUN-07, PIT-RAIN-04	Covered
ECSBF_FR_05	Multi-threaded concurrency engine	Engine thread model, broadcast loop	-	LIT-SUN-16, LIT-SUN-17	PIT-SUN-06	Covered
ECSBF_FR_06	Unified signal propagation to active nodes	Engine broadcast loop	-	LIT-SUN-10, LIT-RAIN-07	PIT-SUN-04	Covered
ECSBF_FR_07	Global Identity Registry maintained	IdentityRegistry class, nodeDB map	UT-SUN-01, UT-SUN-03	LIT-SUN-01	PIT-SUN-02	Covered
ECSBF_FR_08	Session finalization on disconnect	Session::invalidate(), cleanup maps	UT-RAIN-03, UT-RAIN-04	LIT-SUN-14, LIT-SUN-15, LIT-RAIN-09	PIT-SUN-07, PIT-RAIN-04	Covered
ECSBF_FR_09	Network identity resolution	socketToNode mapping in Engine	UT-NetworkUtil	LIT-SUN-10	PIT-SUN-04	Covered
ECSBF_FR_10	Cryptographic protection for data	CryptoUtil::encrypt(), generateKey()	UT-SUN-07, UT-SUN-09, UT-RAIN-06	LIT-SUN-08, LIT-SUN-11	PIT-SUN-05	Covered
ECSBF_FR_11	Payload decryption module	CryptoUtil::decrypt()	UT-SUN-08, UT-RAIN-05	LIT-SUN-09, LIT-RAIN-08	PIT-SUN-05	Covered
ECSBF_FR_12	Multi-level logging (FATAL, INFO, WARNING, DEBUG)	Logger::log(), setLogLevel()	UT-SUN-10, UT-SUN-11, UT-SUN-12, UT-RAIN-07	LIT-SUN-02, LIT-SUN-07, LIT-SUN-18, LIT-RAIN-05, LIT-RAIN-10, LIT-RAIN-11, LIT-RAIN-12, LIT-RAIN-13	PIT-SUN-05	Covered